

A Low-Turnover, Volatility-Budgeted Trading Agent for the ACM ICAIF 2026 Trading Agent Competition

Implementation, methodology, evidence and reproduction guide

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Abstract

This document specifies the trading agent submitted by team *alexleaves* to Competition 99 (ACM ICAIF 2026 Trading Agent Competition). The agent allocates across the fixed 30-stock universe seven times per trading day. Its live decisions come from a single, deliberately simple rule (*approach 1*). It holds an equally weighted long-only stock book whose total size (the *exposure*) targets an annualised volatility of 10% within $[0.5, 0.8]$; the remainder stays in cash. It rebalances only when holdings drift outside no-trade bands. Two further approaches were implemented and evaluated: a pooled LightGBM cross-sectional return model (*approach 2*) and an online exponentially weighted blend of alpha “experts” (*approach 4*). Both showed a small, real predictive signal but did not improve competition outcomes after the 0.1% transaction fee, so they run in *shadow mode*: they are computed and logged every round but never traded. We describe the competition mechanics that motivate this design, all mathematical components, the data pipeline, the backtesting and selection methodology, the results, the live execution system, the test suite, and a step-by-step procedure that lets a third party regenerate every dataset, trained model, experiment table and live decision.

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1 Overview

What the agent does. At each decision round the agent:

1. downloads the latest completed hourly bars for the 30 stocks from Yahoo Finance and validates their freshness;
2. estimates the volatility of an equally weighted portfolio from the last 126 completed trading days;
3. sets the stock exposure $e = \text{clip}(0.10/\hat{\sigma}, 0.5, 0.8)$ and the target $w_i^* = e/30$ for every stock;
4. compares w^* with the current portfolio weights and uploads a new target *only* if the deviation exceeds the no-trade bands ($\|w^* - w\|_1 > 0.20$ or $\|w^* - w\|_\infty > 0.05$); otherwise it uploads nothing, which under the competition rules keeps the current holdings at zero cost.

Approaches 2 and 4 compute alternative *tilted* targets every round; these are written to the decision log for later comparison but are not traded.

Headline evidence. Over 224 historical 15-trading-day windows (official data, 2021–2025) and 115 windows (Yahoo data, mid-2024 to October 2026), each simulated exactly like the Official phase, the production configuration finished on average 2.46th and 2.38th respectively among 17 strategies (itself plus a field of 16 simple competitor strategies). It made about one trade per window, with mean turnover ≈ 0.007 . Its average position stayed between 2.15 and 2.81 in every calendar year, including the 2022 bear market. It stayed within $[2.30, 2.48]$ under every parameter perturbation tested.

2 Competition rules and what they imply

2.1 Rules reproduced by the agent and the simulator

The following rules come from the official starter kit (`docs/rules.md`, `docs/evaluation.md`) and the live schedule served by the organiser backend:

- Universe: 30 U.S. equities in six sectors of five (`universe.json`).
- Each phase starts with USD 1,000,000 in cash in an independent account. Validation runs 8–9 October 2026 (14 rounds); Official runs 12–30 October 2026 (15 days, $K = 105$ rounds).
- Seven rounds per day execute at 09:30, 10:30, $\dots$, 15:30 ET. Upload deadlines are 09:10 and then five minutes before each later execution. A window opens 10 minutes after the previous execution (Round 1 opens at 15:40 ET on the previous trading day).
- A decision is a vector of target weights $w \in \mathbb{R}^{30}$ with $0 \leq w_i \leq 0.30$ and $\sum_i w_i \leq 1$; the remainder is cash. Long-only, no leverage, fractional shares.
- Each rebalance pays a fee of $\phi = 0.1\%$ of total buy-plus-sell notional, including the initial allocation.
- The first upload in a window consumes the round. A missing (or invalid) decision *holds* the portfolio with zero traded notional.

2.2 Evaluation metrics

Order non-cancelled rounds $k = 1, \dots, K$. Let B_k be the NAV immediately before round k executes, $E_k = B_{k+1}$ the NAV at the end of its period, E_K the final official close, N_k the round's buy-plus-sell notional, and $r_k = E_k/B_k - 1$. The four metrics are

$$M_1 = \frac{E_K}{B_1} - 1, \quad M_2 = \sqrt{1764} \frac{\bar{r}}{s_r}, \quad (1)$$

$$M_3 = \max_{1 \leq m \leq L} \frac{P_m - V_m}{P_m}, \quad P_m = \max_{j \leq m} V_j, \quad M_4 = \frac{1}{K} \sum_{k=1}^K \frac{N_k}{B_k}, \quad (2)$$

where s_r is the sample standard deviation of r_k and $V_0, \dots, V_L$ is the chronological sequence of the initial NAV, every period endpoint and every daily 16:00 close. Teams are ranked on each metric (higher M_1, M_2 ; lower M_3, M_4), ties receive average ranks, and the *overall rank score* is the mean of the four ranks (lower is better).

2.3 Implications for strategy design

Proposition 1 (Exposure scaling). *Let a strategy hold a fraction $e \in (0, 1]$ of NAV in a fixed-composition stock portfolio with period returns ρ_k , and the rest in zero-return cash. Ignoring fees, $r_k = e\rho_k$. Hence M_2 does not depend on e , $M_1 \approx e M_1(1)$ to first order, $M_3 \approx e M_3(1)$ to first order, and the initial allocation alone contributes e/K to M_4 .*

Thus, relative to fully invested competitors, lowering exposure improves two ranks (M_3, M_4), leaves one unchanged (M_2) and worsens one (M_1) only when markets rise. The optimal exposure is a property of the field of competitors, not of the market alone (Section 9.2).

Fees against predictability (back-of-envelope). Let an active, dollar-neutral tilt $a = Az/\|z\|_1$ follow standardised signal scores z whose rank correlation (information coefficient, IC) with next-day relative returns is ρ . Let the cross-sectional standard deviation of daily returns be σ_x . For roughly Gaussian z with $N = 30$, $\|z\|_2^2/\|z\|_1 \approx \sqrt{\pi/2}$, so the expected daily active return is approximately

$$\mathbb{E}[a^\top r] \approx \sqrt{\pi/2} \rho \sigma_x A. \quad (3)$$

If the signal renews daily, the active book turns over roughly A per day at cost ϕA . In our data $\sigma_x \approx 1.65\%$ (mean over 2021–2025), so break-even requires $\rho \gtrsim \phi/(\sqrt{\pi/2} \sigma_x) \approx 0.048$. The best model we trained has out-of-sample IC between 0.006 and 0.029 (Table 2). We therefore expect, and observe, that trading on such signals destroys value unless turnover is suppressed so heavily that the tilt is almost never traded.

3 Data

3.1 Official development dataset

The organisers publish hourly OHLCV bars for the 30 stocks from 4 January 2021 to 31 December 2025 as Codabench dataset 1833 (`hourly_market_data_2021_2026.zip`, SHA-256 396a36e8...51652, 263,001 rows, one Parquet file). Bars are timestamped at their *start* in naive U.S./Eastern time and are clock-aligned: 09:30–10:00, 10:00–11:00, ..., 15:00–16:00. The pipeline (`bot/data.py`) verifies the checksum and applies three corrections:

1. **Early closes.** On the ten NYSE 13:00 early-close days, 520 thin after-hours rows (bars after 13:00) are removed. The 13:00 bar, which carries the closing auction, is kept.
2. **Outages.** On five days (2021-04-19, 2021-10-25, 2022-01-24, 2022-01-26, 2022-03-08) most bars are missing. The data is placed on a complete (session bar $\times$ ticker) grid; 469 missing bars are filled flat from the previous close with zero volume and flagged.
3. **Spinoffs.** Prices are split-adjusted but not adjusted for spinoffs, which appear as spurious overnight losses. With pre/post value ratios f (GE 2023-01-04: 1.281; GE 2024-04-02: 1.253; T 2022-04-11: 1.324), each price before a spinoff is divided by the product of all later factors of that ticker: $\tilde{P}_t = P_t / \prod_{s>t} f_s$.

Prices are not dividend-adjusted. This has negligible effect on volatility estimates and does not affect cross-sectional features beyond dividend days.

3.2 Yahoo Finance data

The live agent reads Yahoo Finance through `yfinance` (hourly bars, `prepost=False`, unadjusted for dividends). Yahoo hourly bars start on the half hour: 09:30–10:30, ..., 14:30–15:30, 15:30–

16:00. They therefore coincide exactly with the competition execution times, which is not true of the official bars. Hourly history is limited to about 730 days (the cached file starts on 2023-11-08; the first week, which lacks some tickers, is trimmed, so the panel starts on 2023-11-15). Daily bars since 2021 are also cached for cross-checks. Bars that have not finished at download time are discarded.

Cross-source agreement. After spinoff back-adjustment, daily closes built from the official bars agree with Yahoo daily closes to a median absolute difference of 0.017% (99th percentile 0.149%) over 1,255 days. Only 58 ticker-days differ by more than 0.5%, almost all on the outage days above.

3.3 Bar timing

Let a day's bars start at $s_1 < \dots < s_m$. Bar $j < m$ ends at s_{j+1} . The last bar ends at the session close $c = 13:00$ if $s_m < 15:00$ (early close) and 16:00 otherwise (`bot/market.py`, `bar_end_times`). A bar is *available* at time t if its end is $\leq t$. The execution price at time τ is the open of a bar starting at τ if one exists (always true for Yahoo bars). Otherwise it is the midpoint $(O + C)/2$ of the bar containing τ ; this approximation is used only for the 10:30–15:30 executions on the clock-aligned official data.

4 Decision timeline and information sets

For round k with execution time τ_k , the upload window opens at $o_k = \tau_{k-1} + 10$ min, where τ_{k-1} is the previous executed round (the previous trading day's last round for Round 1). Rounds whose execution would be at or after the session close are cancelled. The live runner decides at the *cutoff*

$$t_k = o_k + 3 \text{ min}, \quad (4)$$

so Yahoo has time to publish the bar that ended at τ_{k-1} . The information set is

$$\mathcal{I}_k = \{\text{bars with end} \leq t_k\}, \quad \mathcal{D}_k = \{\text{days } d : d < \text{date}(t_k)\} \cup \{\text{date}(t_k) \text{ if } t_k \geq 16:00\}, \quad (5)$$

where $\mathcal{D}_k$ is the set of *completed* days whose closes feed the risk model. With Yahoo bars, Round 1 (decided at 15:43 ET the day before) sees data up to the 14:30–15:30 bar. Round 2 (09:43) sees the previous day's close. Round $k \geq 3$ sees the bar ending at τ_{k-1} . The backtester uses exactly the same cutoffs (`build_rounds`), so the simulated information lag matches the live system.

5 Approach 1: volatility-budgeted core allocation (traded)

5.1 Risk model

From the completed daily closes $C_{d,i}$, $d \in \mathcal{D}_k$, take the last $T = 126$ simple returns $x_{d,i} = C_{d,i}/C_{d-1,i} - 1$, forming $X \in \mathbb{R}^{T \times 30}$. The covariance estimate is the Ledoit–Wolf shrinkage estimator

$$\hat{\Sigma} = 252 [(1 - \delta^*)S + \delta^* \mu I], \quad \mu = \frac{1}{N} \text{tr } S, \quad (6)$$

where S is the sample covariance and $\delta^* \in [0, 1]$ the analytically optimal shrinkage intensity (scikit-learn `LedoitWolf`). An exponentially weighted alternative (half-life 30 days) was tested for robustness. If fewer than $\max(30, T/2)$ returns are available, the agent falls back to equal weights at the configured maximum exposure.

5.2 Portfolio constructions considered

All constructions produce weights $u \geq 0$ with $\sum_i u_i = 1$, subject to a per-stock cap $\bar{u}$ and a sector cap $\bar{g} = 0.30$. They are then scaled to the exposure e .

Equal weight (production).

$$u_i = 1/30.$$

Inverse volatility.

$$u_i \propto \hat{\sigma}_i^{-1} \text{ with } \hat{\sigma}_i = \sqrt{\hat{\Sigma}_{ii}}.$$

Risk parity (equal risk contribution).

Solve $y^* = \arg \min_{y>0} \frac{1}{2} y^\top \hat{\Sigma} y - \frac{1}{N} \sum_i \log y_i$ (L-BFGS-B). The first-order conditions give $y_i(\hat{\Sigma} y)_i = 1/N$, so every asset contributes equally to the variance of y . Then $u = y^* / \sum y^*$.

Minimum variance.

$\min_u u^\top \hat{\Sigma} u$ subject to $\sum u_i = 1$, $0 \leq u_i \leq \bar{u}$ and $\sum_{i \in g} u_i \leq \bar{g}$ for every sector g (SLSQP, analytic gradients).

Cap enforcement. The operator $\text{cap}(v, \text{total}, \bar{u}, \bar{g})$ (`risk.cap_weights`) rescales v to the requested total. It then repeatedly clips names above $\bar{u}$ and sectors above $\bar{g}$ and redistributes the excess proportionally to unconstrained names and sectors. Each pass saturates at least one asset or group, so the loop terminates. For equal weights at $e \leq 0.8$ no cap binds ($0.8/30 \approx 0.027$).

5.3 Volatility-targeted exposure

With $\hat{\sigma}_p = \sqrt{u^\top \hat{\Sigma} u}$ the predicted annualised volatility of the unit-sum portfolio,

$$e = \text{clip}\left(\frac{\sigma^*}{\hat{\sigma}_p}, e_{\min}, e_{\max}\right), \quad \sigma^* = 0.10, \quad e_{\min} = 0.5, \quad e_{\max} = 0.8, \quad w^* = e u. \quad (7)$$

For equal weights $\hat{\sigma}_p = \frac{1}{30} \sqrt{\mathbf{1}^\top \hat{\Sigma} \mathbf{1}}$. Over 2021–2025 the realised volatility of the equally weighted portfolio was 17.3% p.a. (so $e \approx 0.58$ on average). On 6 October 2026 the estimate was 9.98%, giving $e = 0.8$. Volatility targeting lowers exposure when risk rises, which protects M_3 precisely when drawdowns are most likely. The target is recomputed once per completed day, so it changes slowly.

5.4 Current weights and the no-trade rule

At the cutoff, with share holdings s , cash c and latest available closes P , the current weights are $w_i = s_i P_i / (c + \sum_j s_j P_j)$. Live, s and c come from the backend portfolio endpoint and P from Yahoo (Section 10). With deviation $\Delta = w^* - w$, the agent trades if and only if

$$\|\Delta\|_1 > b_1 = 0.20 \quad \text{or} \quad \|\Delta\|_\infty > b_\infty = 0.05. \quad (8)$$

When trading, the new target is $w + \kappa \Delta$ with $\kappa = 1$ (full move to target). It is clipped to $[0, 0.30]$, sector caps are re-imposed and the total is kept at most 1 (`agent.finalize`). Otherwise nothing is uploaded and the portfolio is held. Because an equal-weight book drifts slowly, in practice the agent trades at the first round of a phase and afterwards only when the volatility-targeted exposure moves by more than about 0.2 in total (about 1.0–1.3 trades per 15-day window in backtests).

Upload format. Weights are rounded *down* to six decimals as exact decimal numbers (function `quantize` in `bot/runner.py`). If rounding ever left the sum above 1, the largest weight would be reduced by 10^{-6} until it is not. The payload contains all 30 symbols and passes the kit’s own validator.

Parameter	Value	Meaning
construction	equal_weight	unit-sum composition u
cov_method	Ledoit-Wolf	covariance estimator
lookback_days	126	completed daily returns used
vol_target	0.10	σ^*
min_exposure, exposure	0.5, 0.8	$e_{\min}, e_{\max}$
cap, sector_cap	0.10, 0.30	per-stock and per-sector caps (non-binding for equal weight)
band_l1, band_max	0.20, 0.05	no-trade bands b_1, b_{∞}
trade_fraction	1.0	κ

Table 1: Production parameters of approach 1 (bot/production.json).

6 Approach 2: pooled LightGBM cross-sectional model (shadow)

6.1 Features

Features are computed from the bar panel at the last bar available at the cutoff (bot/features.py). Every feature uses only rolling windows, backward shifts and same-row cross-sectional operations. Hence features computed on a truncated history equal those computed on the full history at the same bar; this is enforced by a unit test. With C, H, L, V the close, high, low and volume panels, $\ell_t = \log C_t - \log C_{t-1}$, and $r_t^{(k)} = C_t/C_{t-k} - 1$:

Feature (count)	Definition
rel_rk, rank_rk (14)	$r^{(k)}$ minus its cross-sectional mean; cross-sectional percentile rank of $r^{(k)}$ minus $\frac{1}{2}$; $k \in \{1, 3, 7, 14, 35, 70, 140\}$ bars
volk (3)	rolling standard deviation of ℓ over $k \in \{7, 35, 140\}$ bars (minimum $\max(3, k/2)$ observations)
rank_vol35 (1)	cross-sectional rank of vol35
range1, range7 (2)	$(H - L)/C$ and its 7-bar mean
dist_high35, dist_low35 (2)	$C/\max_{35} H - 1$ and $C/\min_{35} L - 1$
volume_z, rank_volume_z (2)	$\log(1 + V_t) - \log(1 + \bar{V})$ with $\bar{V}$ the mean volume of the previous 20 bars in the same intraday slot; its rank
rel_intraday (1)	$C_t/C_{\text{prev. day}}^{\text{close}} - 1$, cross-sectionally demeaned
market (6)	7- and 35-bar sums of the equal-weight mean log return; 35-bar volatility of that mean; cross-sectional std of $r^{(7)}$; intraday slot index; weekday

That gives 31 features in total. Horizons are expressed in bars, so a 7-bar window is about one trading day.

6.2 Labels

For each round k , the forward return is measured from execution to execution over h rounds, $R_{k,i} = P_{\tau_{k+h},i}/P_{\tau_k,i} - 1$. The label is its cross-sectional percentile rank, centred:

$$y_{k,i} = \text{rank}_i(R_{k,i} - \frac{1}{N} \sum_j R_{k,j})/N - \frac{1}{2}. \quad (9)$$

We tested $h = 7$ (one day) and $h = 35$ (one week). One training row is a (round, ticker) pair. Pooling all tickers in one model multiplies the sample size by 30 and discourages ticker-specific overfitting.

6.3 Model and training

LightGBM gradient-boosted trees with the deliberately conservative hyperparameters below, fixed *a priori* and not tuned:

objective=regression, learning_rate=0.03, num_leaves=15, max_depth=4,
min_data_in_leaf=500, feature_fraction=0.8, bagging_fraction=0.8, bagging_freq=1,
lambda_l2=10, num_boost_round=300, seed=7, deterministic=true, num_threads=8.

Walk-forward protocol with purging. Training uses an expanding window starting on 1 January 2021. The last h rounds before each training end are dropped, so no label overlaps the test period. Folds (test periods): A = 2023H1, B = 2023H2, C = 2024H1, D = 2024H2, holdout = 2025. The production model is retrained on all official data (2021–2025, 258,180 rows) and evaluated on Yahoo data from January to October 2026. This is a genuinely unseen period, with bars aligned differently from the training data.

6.4 Predictive results

The information coefficient of round k is the Spearman correlation between scores and forward relative returns. The t -statistic accounts for the overlap of h -round labels: $t = \overline{IC}/s_{IC} \cdot \sqrt{n/h}$.

Test period (rounds)	$h = 7$ (1 day)			$h = 35$ (1 week)		
	mean IC	t	hit rate	mean IC	t	hit rate
A: 2023H1 (868)	0.006	0.31	0.50	−0.041	−0.94	0.41
B: 2023H2 (876)	0.025	1.31	0.55	0.019	0.48	0.57
C: 2024H1 (868)	0.024	1.26	0.56	−0.038	−0.98	0.43
D: 2024H2 (887)	0.012	0.63	0.53	0.013	0.30	0.54
Holdout: 2025 (1740)	0.012	0.86	0.51	−0.010	−0.32	0.46
Yahoo 2026 (1336)	0.029	1.94	0.56	0.021	—	0.52

Table 2: Out-of-sample information coefficients of the LightGBM model. The one-day model is positive in every period; the one-week model is not and was discarded.

For comparison, single-feature signals on 2023–2025 had mean IC of 0.007 (5-day momentum), 0.002 (20-day momentum), 0.004 (1-day reversal) and −0.016 (low volatility), and their signs flip in 2026. The model’s highest gain shares are market volatility (0.12), stock volatility over 35 and 140 bars (0.09, 0.08), the 35-bar market return (0.08) and 10–20-day relative returns. The model therefore mostly learns how the cross-section of volatile versus quiet stocks behaves in different market regimes. The signal is real but, at $IC \leq 0.03$, below the break-even estimate of Section 2.3.

6.5 Tilt operator

Model scores m are standardised cross-sectionally ($z = \text{clip}((m - \bar{m})/s_m, -3, 3)$; missing values set to 0). They then tilt the core weights multiplicatively at constant exposure:

$$w^{\text{tilt}} = \text{cap}\left(w^* \odot \max(0, 1 + \lambda \text{clip}(z, -c, c)), \sum_i w_i^*, \bar{u}, \bar{g}\right), \quad c = 2. \quad (10)$$

Strengths $\lambda \in \{0.15, 0.3, 0.6\}$ were evaluated; the shadow log uses $\lambda = 0.15$.

7 Approach 4: online expert weighting (shadow)

Five experts produce raw signals at each cutoff: *momentum* (`rel_r35`), *long momentum* (`rel_r140`), *reversal* (`−rel_r7`), *low volatility* (`−vol35`) and the LightGBM *model*. Each is standardised to $z_k^{(n)}$. Between consecutive decisions $k - 1$ and k , with stock returns $r_k = P_k/P_{k-1} - 1$ measured on the latest closes at the two cutoffs, expert n earns the return of its dollar-neutral portfolio:

$$G_k^{(n)} = \frac{z_{k-1}^{(n)\top} r_k}{\|z_{k-1}^{(n)}\|_1}, \quad S_k^{(n)} = \gamma S_{k-1}^{(n)} + G_k^{(n)}, \quad \gamma = 0.995 \quad (\text{half-life} \approx 138 \text{ rounds}). \quad (11)$$

The expert weights follow an exponentiated-gradient (Hedge) proposal with capped steps:

$$p_k = \text{softmax}(\eta S_k), \quad \eta = 100; \quad q_k = \frac{\max(q_{k-1} + \text{clip}(p_k - q_{k-1}, -\delta, \delta), 10^{-6})}{\|\max(\cdot)\|_1}, \quad \delta = 0.05. \quad (12)$$

Forgetting ($\gamma < 1$) lets the weights track regime changes. The step cap δ prevents a few noisy hours from replacing a historically reliable expert. The blended score $\alpha_k = \sum_n q_k^{(n)} z_k^{(n)}$ feeds the tilt operator above ($\lambda = 0.15$ in the shadow log). The state (S , q , the last signals and prices) is a small JSON object, persisted by the live runner between rounds. The expert weights depend only on market data, not on the portfolio, so in backtests they are computed in one continuous pass.

8 Backtesting and selection methodology

8.1 Simulator

`bot/backtest.py` replays the competition round by round. At round k with execution prices P_k , holdings s and cash c : $B_k = c + s^\top P_k$. If the strategy returns a target w , the holdings value v and the fee f solve the fixed point

$$v = w(B_k - f), \quad f = \phi \|v - s \odot P_k\|_1, \quad (13)$$

solved by four iterations from $f = 0$ (it converges to machine precision because $\phi \ll 1$). Then $s \leftarrow v/P_k$, $c \leftarrow B_k - f - \mathbf{1}^\top v$ and $N_k = \|v - s_{\text{old}} \odot P_k\|_1$. A hold sets $N_k = 0$. Current weights for the decision are marked at the latest close available at the cutoff, as in live trading. Period endpoints, daily 16:00 closes and the final close form the valuation sequence. The metric implementation agrees with the kit’s reference calculator (`kit/evaluation.py`, 40-digit decimals) to relative error 10^{-9} (unit test). The backend’s exact fee convention (whether targets are sized on pre- or post-fee NAV) is not published. Our convention differs from the alternative by second-order terms (ϕ^2).

8.2 Windows

Every trading day starts a candidate window, with a stride of 5 days. A window is 15 trading days (105 rounds, the Official phase length) and starts from USD 1,000,000 in cash, so it includes the initial allocation fee. Official data provides 224 windows from July 2021 (after a 126-day warm-up). Yahoo data provides 115 windows from June 2024.

8.3 Competitor field and rank scoring

Since the score is a rank, we score each candidate against a field of 16 simple strategies that plausibly bracket real teams:

- equal weight bought and held (`ew_hold`), equal weight rebalanced daily (`ew_daily`), and 50% equal weight with 50% cash (`half_cash_ew`);
- top-5 momentum rebalanced every round (`momentum_top5_hourly`, close to the kit’s example agent) and daily (`momentum_top5_daily`), and top-5 short-term reversal every round;
- the 10 lowest-volatility stocks daily, and the top-4 20-day trend stocks held;
- eight random equal-weight portfolios of 5–12 stocks redrawn daily (`seed=11`), proxies for noisy agents.

In each window, each of the four metrics is ranked across the field plus the candidate (average ranks for ties). The overall rank score is their mean, and the *position* is 1+ the number of field members with a strictly lower overall score. Because the field is invented, positions are also computed against a *sensible* sub-field of six members (no random or every-round strategies). The selection criterion is the mean, over both datasets, of $\frac{1}{2} \left[\frac{\text{pos}-1}{16} + \frac{\text{pos}_{\text{sens}}-1}{6} \right]$, i.e. the normalised expected finishing position.

8.4 Search space and gate

Approach 1 was searched on a grid of $4 \times 2 \times 3 \times 4 = 96$ configurations, with per-stock cap 0.10 and lookback 126:

- construction: equal weight, inverse volatility, risk parity, minimum variance;
- volatility target: none or 0.10;
- maximum exposure: 0.6, 0.8 or 1.0;
- bands (b_1, b_∞) : (0.05, 0.02), (0.10, 0.03), (0.20, 0.05) or (0.40, 0.10).

Approaches 2 and 4 were evaluated on top of the chosen core, with $\lambda \in \{0.15, 0.3, 0.6\}$ and bands (0.20, 0.05) and (0.40, 0.10). For approach 2, every backtest round uses scores from the walk-forward model whose training ended before that round (folds A–D, holdout, then the final model for 2026). The **gate** for trading an overlay was that it must improve the expected position on both datasets after costs.

9 Results

All returns and drawdowns are per 15-day window, in percent; turnover is M_4 ; “pos.” is the mean position in the 17-strategy field.

9.1 The competitor field

Strategy	Official data, 2021–2025 (224 windows)					Yahoo data, 2024–2026 (115 windows)				
	ret	Sharpe	MDD	turnover	pos.	ret	Sharpe	MDD	turnover	pos.
half_cash_ew	0.34	1.55	1.99	0.0048	2.08	0.54	1.95	1.56	0.0048	1.91
ew_hold	0.68	1.58	3.94	0.0095	3.08	1.08	1.98	3.10	0.0096	2.95
ew_daily	0.66	1.57	3.96	0.0111	3.88	1.06	1.97	3.12	0.0113	3.60
trend_top4_hold	1.43	1.39	5.40	0.0095	5.33	1.23	1.02	5.44	0.0096	6.03
low_vol_top10_daily	−0.07	0.33	3.49	0.0559	5.39	0.18	0.43	3.04	0.0588	5.68
momentum_top5_daily	0.24	0.19	5.58	0.1115	7.46	0.58	0.50	5.41	0.1108	7.58
random (mean of 8)	−1.23	−1.08	5.20	0.1984	9.67	−1.01	−0.95	4.63	0.2014	9.72
momentum_top5_hourly	−5.28	−5.01	8.35	0.5831	14.34	−3.33	−2.76	7.45	0.5477	13.41
reversal_top5_hourly	−9.38	−9.04	11.08	0.9878	15.75	−9.01	−7.49	10.58	0.9711	15.62

Table 3: Field members ranked against each other. Strategies that trade every round lose 3–9% per window to fees. The best field member is the half-cash equal-weight portfolio, which illustrates Proposition 1.

9.2 Approach 1: grid results and the choice of configuration

Table 4 averages both datasets at bands (0.20, 0.05). Equal weight dominates the other constructions on expected position and top-3 frequency. Minimum variance is the worst: it concentrates in a few low-volatility names, which lowers Sharpe without lowering drawdown enough. Lower maximum exposure helps, as Proposition 1 predicts, and volatility targeting helps slightly. Band width matters little (Table 5) because an equal-weight book rarely drifts beyond any of the tested bands.

Judgement in the final choice. The grid winner under the selection criterion is equal weight, volatility target 0.10, $e_{\max} = 0.6$, bands (0.40, 0.10). We deliberately chose $e_{\max} = 0.8$ and bands (0.20, 0.05) instead, for two reasons. First, the advantage of 0.6 (0.07 positions) comes from the drawdown and turnover ranks against a field that is *assumed*; if real competitors earn positive returns more often than our field does, the return rank penalty of a smaller book grows. Second, the narrower band keeps the ability to cut exposure when volatility jumps, while costing only 0.05 positions. Both alternatives are one-line changes to `bot/production.json`.

Construction	vol. target	$e_{\max}$	ret	Sharpe	MDD	turnover	pos.	pos. (sensible)	top-3 %
equal weight	0.10	0.6	0.44	1.69	2.09	0.0054	2.36	2.02	89
equal weight	none	0.6	0.49	1.69	2.21	0.0057	2.41	2.12	88
equal weight	0.10	0.8	0.50	1.68	2.54	0.0068	2.43	2.14	86
equal weight	0.10	1.0	0.52	1.69	2.73	0.0073	2.49	2.18	83
equal weight	none	0.8	0.65	1.70	2.94	0.0076	2.42	2.27	85
inverse volatility	0.10	0.6	0.38	1.65	1.92	0.0055	2.65	2.27	70
inverse volatility	0.10	0.8	0.44	1.64	2.36	0.0068	2.67	2.37	70
risk parity	0.10	0.6	0.39	1.69	1.83	0.0056	2.78	2.36	64
risk parity	0.10	0.8	0.44	1.69	2.25	0.0070	2.80	2.47	64
equal weight	none	1.0	0.82	1.71	3.66	0.0096	3.13	2.96	61
risk parity	none	1.0	0.67	1.71	3.13	0.0096	3.31	3.06	53
min variance	0.10	0.6	0.31	1.39	1.70	0.0059	3.23	2.55	57
min variance	none	1.0	0.51	1.39	2.83	0.0109	4.10	3.44	47

Table 4: Selected rows of the 96-configuration grid (both datasets pooled, bands (0.20, 0.05)). The production configuration is in bold. The full grid is in `private/results/core_windows.csv`.

b_1 (b_∞)	trades / window	turnover	ret	pos.	pos. (sensible)
0.05 (0.02)	1.30	0.0069	0.49	2.51	2.22
0.10 (0.03)	1.06	0.0068	0.48	2.45	2.15
0.20 (0.05)	1.04	0.0068	0.50	2.43	2.14
0.40 (0.10)	1.00	0.0067	0.50	2.38	2.09

Table 5: Effect of the no-trade bands on the production configuration.

9.3 Approaches 2 and 4

Data	Target	λ	ret	Sharpe	MDD	turnover	trades	pos.
official	core (approach 1)	–	0.75	2.38	2.23	0.0068	1.0	2.44
	LightGBM tilt (2)	0.15	0.75	2.40	2.23	0.0068	1.0	2.57
		0.30	0.66	2.26	2.29	0.0164	5.7	3.78
		0.60	0.15	1.14	2.50	0.0673	26.5	5.86
	online experts (4)	0.15	0.75	2.38	2.23	0.0068	1.0	2.66
		0.30	0.71	2.32	2.25	0.0093	2.2	3.19
		0.60	0.48	1.60	2.37	0.0260	9.4	4.79
Yahoo	core (approach 1)	–	0.67	1.96	2.34	0.0071	1.1	2.38
	LightGBM tilt (2)	0.15	0.68	1.98	2.35	0.0071	1.1	2.73
		0.30	0.49	1.71	2.43	0.0219	8.1	4.02
		0.60	–0.13	0.55	2.69	0.0793	29.7	6.10
	online experts (4)	0.15	0.66	1.94	2.34	0.0071	1.1	2.70
		0.30	0.60	1.83	2.33	0.0107	2.8	3.20
		0.60	0.24	1.07	2.45	0.0328	11.6	5.30

Table 6: Overlays on the production core, bands (0.20, 0.05), on periods with out-of-sample model scores (official 2023–2025, Yahoo June 2024–October 2026).

Table 6 confirms the fee argument. Whenever a tilt is strong enough to trigger trades after the initial allocation, turnover multiplies and every metric worsens. At $\lambda = 0.15$ the tilt only shapes the initial allocation (1.0 trades per window) and yields statistically indistinguishable return and Sharpe, but a slightly worse position (2.57 vs. 2.44; 2.73 vs. 2.38). With wider bands (0.40, 0.10) the picture is the same: the best tilt reaches position 2.51 (official) and 2.64 (Yahoo) against 2.38 and 2.33 for the core. Neither overlay passes the gate, so both remain in shadow mode. The learned expert weights also differ sharply between the two datasets (long momentum dominant in one, reversal and momentum in the other), further evidence of an unstable edge.

9.4 Robustness

Variant	Official 2021–2025				Yahoo 2024–2026			
	ret	MDD	pos. (fee 0.1%)	pos. (fee 0.2%)	ret	MDD	pos. (0.1%)	pos. (0.2%)
production	0.40	2.64	2.46	2.01	0.67	2.34	2.38	2.12
lookback 63 days	0.47	2.65	2.41	2.01	0.74	2.37	2.34	2.12
lookback 252 days	0.39	2.58	2.44	1.99	0.66	2.22	2.30	2.12
EWMA covariance	0.44	2.56	2.38	1.96	0.69	2.27	2.36	2.03
$e_{\max} = 0.7$	0.39	2.44	2.44	1.94	0.63	2.08	2.30	2.04
$e_{\max} = 0.9$	0.43	2.75	2.48	2.01	0.70	2.51	2.37	2.13
bands $\times 0.5$	0.39	2.64	2.47	2.02	0.65	2.35	2.41	2.14
bands $\times 2$	0.41	2.63	2.41	1.98	0.68	2.33	2.33	2.05

Table 7: Perturbations of the production configuration. Changing the per-stock cap ($\times 0.75$, $\times 1.5$) or blending with equal weight gives results identical to production because no cap binds for an equal-weight book. Doubling the fee *improves* the relative position because high-turnover competitors are hurt more.

Data	Year	windows	ret	Sharpe	MDD	turnover	pos.
official	2021	26	0.57	1.56	2.61	0.0076	2.15
	2022	50	−0.56	−0.63	3.81	0.0053	2.64
	2023	50	0.80	2.17	1.97	0.0064	2.22
	2024	51	0.93	2.55	2.13	0.0076	2.33
	2025	47	0.34	2.09	2.67	0.0064	2.81
Yahoo	2024	30	0.88	2.03	2.42	0.0077	2.17
	2025	50	0.48	2.13	2.57	0.0065	2.64
	2026	35	0.77	1.64	1.97	0.0076	2.20

Table 8: Production configuration by window start year. In the 2022 bear market the strategy loses money but keeps its relative position, because lower exposure and near-zero turnover protect drawdown and turnover ranks.

10 Live execution system

The runner (`python -m bot.runner -phase validation|official`) replaces the kit’s `watch` command, which cannot skip a round, stops entirely if the strategy raises, and holds the checkpoint lock for its whole lifetime. The runner opens a kit session only for each platform interaction, so the kit’s other commands remain usable while it runs.

Data validation. A download is rejected (and retried) if any symbol is missing or has a non-positive price. It is also rejected if the last bar ends before the latest bar end $\hat{e}(t)$ that Yahoo should have published. $\hat{e}(t)$ is the latest of today’s 10:30, ..., 15:30, 16:00 boundaries at or before $t - 2$ min on weekdays, otherwise the previous weekday’s 16:00. On a market holiday this check fails safe (the round is skipped and the portfolio held). Up to three symbols may miss only the latest bar, because Yahoo occasionally publishes a final bar late; this was observed for TMO on 6 October 2026. Their previous close is carried forward and they are listed in the log.

Failure handling.

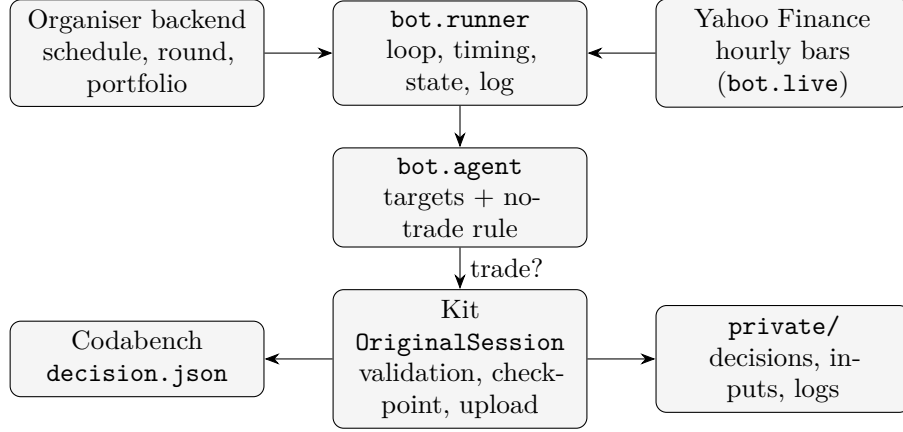


Figure 1: Live architecture. All platform calls go through the starter kit’s own client.

Algorithm 1: Runner loop for one phase

```

1 while phase not complete do
2   read schedule (server clock is authoritative);
3   if a round  $k$  is open and not yet handled then
4     wait until  $o_k + 3$  min;
5     if a decision file for  $k$  exists then re-upload it verbatim (resume) and stop;
6     if the round already has an eligible attempt then mark occupied;
7     read portfolio;
8     fetch and validate bars; retry every 2 min until deadline – 10 min, else mark skipped (hold);
9     compute targets (approach 1 traded; 2 and 4 logged) and the no-trade rule;
10    if no trade then mark held; upload nothing;
11    else
12      write decision.json and inputs.parquet; upload through
13      OriginalSession.decision; mark submitted
14    append a log record; persist state;
  sleep until the next window opens (at most 5 min);

```

Situation	Behaviour
Data stale or missing until the deadline	round skipped; holdings kept; no fee
Exception inside the agent	round skipped; loop continues
Platform error before the decision file exists	retried at the next poll (5 min)
Interrupted upload after the file exists	the identical bytes are resubmitted; the kit check-point reconciles by SHA-256 and never creates a duplicate
Ambiguous creation that the kit cannot reconcile	no automatic retry (kit policy); logged as <i>error</i> , then <i>missed</i> after the deadline
Slot already used	marked <i>occupied</i> ; nothing uploaded
Backend positions in an unknown format	falls back to the backend weight snapshot; if that is empty, the round is skipped rather than mistaking holdings for cash

Audit trail. Every round appends one JSON line to `private/logs/decisions.jsonl` containing the code version (git commit), the data diagnostics, the current weights and how they were obtained, all three targets (approach 1, 2, 4), the expert weights, the deviation, the decision and the platform receipt. The team token is never logged. For every submitted round the exact uploaded `decision.json` and the bar panel the agent saw (`inputs.parquet`) are archived under `private/decisions/<phase>/<round>/`.

11 Verification

The suite (`python -m pytest tests`, 65 tests, about 5 seconds) runs on synthetic data and a fake platform, so it needs neither credentials nor network:

- **Constraints:** every construction respects stock and sector caps and its total, for random covariance matrices. Quantised weights always pass the kit's `_weights` validator, including NaN inputs.
- **No lookahead:** features are invariant to truncation of the history. Multiplying all prices after a cutoff by random factors leaves every target at that cutoff unchanged, both live-style and in the backtester.
- **Accounting:** simulator metrics equal the kit's decimal reference calculator. Initial-allocation fee and turnover are exact. An all-cash strategy has zero return, drawdown and turnover.
- **End-to-end:** an in-memory Codabench and backend (`httpx.MockTransport`) drives the real kit client. Tests check one valid upload per round, holds within bands, skips on data or agent failure, respect for occupied slots, that a dry run never uploads, verbatim resume of an existing decision file, no duplicate after an ambiguous upload failure, and a complete simulated trading day (one trade, six holds, phase completion).
- **Replay:** a submitted decision is recomputed bit-for-bit from its archived inputs (`bot.replay`).

In addition, a dry run against the live organiser API (read-only) on 6 October 2026 produced a decision for `validation-2026-10-08-r1` that the kit validator accepted for an upload time inside that round's window.

12 Reproduction guide

12.1 Environment

Linux (tested on kernel 5.15), Python 3.13.2. Exact package versions are pinned in `requirements-bot.txt`: `lightgbm 4.7.0`, `numpy 2.5.3`, `pandas 3.0.6`, `pyarrow 25.0.1`, `scikit-learn 1.9.1`, `scipy 1.18.1`, `yfinance 1.7.0`, `httpx 0.28.1`, `pytest 9.1.1`. No GPU is needed. The full experiment study takes about 12 minutes on 20 CPU cores; model training takes about 10 seconds.

12.2 Steps

All commands run from the `starter-kit` directory.

1. Install.

```
python3.13 -m venv .venv && .venv/bin/activate
pip install -r requirements-bot.txt
cp .env.example .env && chmod 600 .env
# in .env: CODABENCH_TOKEN=<personal token>
#          ICAIF_PROFILE=profiles/profile99-production.json
```

2. Build the data (downloads dataset 1833, verifies its SHA-256, cleans it, downloads Yahoo data, writes `private/data/data_report.json`).

```
python -m bot.data --refresh
```

3. Run the tests.

```
python -m pytest tests
```

4. Reproduce the study and the model. The first command runs the 96-configuration grid and reports the grid winner. The second evaluates overlays and robustness on the production

core recorded in `bot/production.json` (field `selected_by`). Both commands train the walk-forward models and save the production model to `bot/models/lgbm.txt` with metadata `lgbm.json`.

```
python -m bot.experiments
python -m bot.experiments --core '{"core": "equal_weight|cap0.1|vt0.1|ex0.8|lb126|blend0.0", "band": {"band_l1": 0.2, "band_max": 0.05}, "target": "base", "alpha": null, "online": null}'
sha256sum bot/models/lgbm.txt
# expected: 0fe5e29dca33147f92af90df10de88eb
#           6eaef4b67007836d388ec8895fc3650e
```

Outputs go to `private/results/`: a summary (`experiments.json`) and one CSV file of per-window results for each stage. Every table in this document was produced from these files.

5. Check a live decision without uploading.

```
python -m bot.runner --phase validation --dry-run --simulate-round validation
-2026-10-08-r1
```

6. Trade a phase. The runner is meant to run unattended; a systemd user unit is provided in `deploy/`.

```
python -m bot.runner --phase validation
python -m bot.runner --phase official
```

7. Regenerate a submitted decision from its archived inputs and logged portfolio. The command exits with status 0 only if the recomputed weights equal the uploaded ones exactly.

```
python -m bot.replay private/decisions/official/official-2026-10-12-r1
```

12.3 Determinism

- The official dataset is fixed by checksum. Cleaning is deterministic.
- LightGBM runs with `deterministic=true`, a fixed seed and a fixed thread count. Retraining on 6 October 2026 reproduced the stored model byte for byte (same SHA-256). The production decision does not use the model.
- The competitor field is generated with a fixed seed (`seed=11`).
- Yahoo data is not under our control: Yahoo may revise bars and keeps only about 730 days of hourly history. Experiments on Yahoo data are therefore reproducible to within such revisions. Live decisions are reproducible exactly from the archived `inputs.parquet` files.
- The production decision depends only on the bar history, the backend portfolio and `production.json`. The shadow targets also depend on the online expert state, which is stored in `private/state/runner_<phase>.json` and logged every round.

12.4 Code map

Path	Role
bot/data.py	dataset download, checksum, cleaning, spinoff adjustment, Yahoo cache, cross-checks
bot/market.py	bar panels, bar end times, round timeline, execution prices
bot/features.py	31 causal features
bot/risk.py	covariance estimators, portfolio constructions, cap enforcement
bot/agent.py	approaches 1, 2 and 4; configuration; no-trade rule
bot/alpha.py	LightGBM dataset, training, purged walk-forward, IC
bot/backtest.py	competition simulator, windows, competitor field, rank scoring
bot/experiments.py	the full study and model export
bot/live.py	live Yahoo download, freshness checks, portfolio parsing
bot/runner.py	live loop and uploads through the kit client
bot/replay.py	bit-exact regeneration of submitted decisions
bot/production.json	production configuration
bot/models/lgbm.{txt,json}	production LightGBM model and metadata
tests/	65 automated tests
deploy/icaif-runner@.service	optional systemd user unit

13 Limitations

- **Invented field.** The rank-based results depend on a field of competitors we constructed. Real teams may be stronger or weaker on return, which changes the optimal exposure (Proposition 1). This is why we report the sensible sub-field as well and chose $e_{\max} = 0.8$ over the marginally better 0.6.
- **Fee convention.** The backend’s exact fee sizing is unpublished; the effect is second order.
- **Execution prices on official data** are approximated by bar midpoints for 10:30–15:30. The Yahoo backtest uses exact bar opens.
- **Short horizon.** A 15-day phase is short. Even a sound strategy can finish poorly in a given window; the worst historical window lost 8.6% with a 13.4% drawdown.
- **Portfolio format.** The backend’s position schema was unknown until the first Validation execution. The parser accepts the common forms and fails safe.
- **Data provider.** The agent depends on Yahoo Finance availability. Outages lead to held rounds, not to erroneous trades.